

Healthcare Analytics Project Report

Project Overview

This Healthcare Analytics project was developed to analyze hospital operations using SQL. The objective was to transform raw healthcare data into meaningful business insights that support operational efficiency, resource planning, patient management, and revenue optimization.

The analysis covers five major business areas:

- Patient Analysis
- Doctor Performance Analysis
- Appointment Analysis
- Treatment Analysis
- Billing & Revenue Analysis

The project demonstrates the use of SQL for data exploration, aggregation, segmentation, ranking, and business intelligence reporting.

Dataset Description

The project consists of five relational tables representing different hospital operations.

Patients

- Patient ID
- Name
- Age
- Gender
- Address
- Registration Date
- Email

Doctors

- Doctor ID
- Doctor Name
- Specialization
- Experience

Appointments

- Appointment ID
- Patient ID

- Doctor ID
- Appointment Date
- Status

Treatments

- Treatment ID
- Treatment Name
- Cost

Billing

- Bill ID
- Patient ID
- Treatment ID
- Amount
- Payment Status

Project Objectives

The primary objectives of this project are:

- Analyze patient demographics and registration trends.
- Evaluate doctor experience and workload.
- Study appointment patterns and operational efficiency.
- Identify the most frequently performed treatments.
- Analyze treatment costs and revenue generation.
- Understand billing performance and payment behavior.
- Generate business recommendations to improve hospital operations.

SQL Analysis & Business Insights

1. Exploratory Data Analysis (EDA)

Total Patients

Observation The hospital database contains **50 registered patients**.

Business Insight The current dataset represents a small patient population suitable for understanding operational trends and testing analytical workflows.

Recommendation Continue monitoring patient growth over time and compare monthly registrations to evaluate hospital expansion.

Total Doctors

Observation The hospital currently has **10 doctors**.

Business Insight This provides the available medical workforce responsible for serving all patient appointments.

Recommendation Regularly compare doctor availability against patient demand to avoid understaffing.

Available Medical Specializations

- Dermatology
- Pediatrics
- Oncology

Business Insight The hospital currently offers services across three major medical specialties.

Recommendation Monitor demand by specialization and consider introducing additional specialties if patient demand continues to grow.

2. Patient Analysis

Recently Registered Patients

Observation Only **1 patient** registered during the last 30 days.

Business Insight Recent patient acquisition is relatively low, which may indicate reduced hospital outreach or lower patient inflow.

Recommendation Launch awareness campaigns, referral programs, and digital marketing initiatives to increase new patient registrations.

Patient Distribution by Address

Address	Patients
321 Maple Dr	19
789 Pine Rd	15
123 Elm St	9

Address	Patients
456 Oak Ave	7

Business Insight Patient concentration is highest in **321 Maple Drive**, indicating that this area contributes the largest share of hospital visitors.

Recommendation Focus local healthcare campaigns and community outreach programs in high-density areas while improving visibility in low-coverage regions.

Patient Age Segmentation

Age Group	Patients
Adults	18
Mature	17
Seniors	15

Business Insight The patient population is evenly distributed across all age groups, indicating balanced healthcare demand.

Recommendation Maintain healthcare services tailored for each age segment, including preventive care for adults and specialized support for senior patients.

Most Common Email Domain

Observation The most frequently used email domain is **mail.com**.

Business Insight Most patients use the same email provider, making email communication a practical engagement channel.

Recommendation Use automated email notifications for appointment reminders, billing updates, and health awareness campaigns.

3. Doctor Analysis

Doctor Experience

Observation

- David Jones has the highest experience with **28 years**.
- **7 doctors** have more than **20 years** of experience.
- **3 doctors** have approximately **3 years** of experience.

Business Insight The hospital has a highly experienced medical workforce supported by a small group of junior doctors.

Recommendation Implement mentorship programs where senior doctors guide junior doctors to improve knowledge sharing and clinical performance.

Specialization Demand

Specialization	Appointments
Pediatrics	98
Dermatology	70
Oncology	32

Business Insight Pediatrics has the highest appointment demand, followed by Dermatology, while Oncology receives comparatively fewer appointments.

Recommendation Allocate additional consultation slots and resources to Pediatrics to reduce patient waiting time.

Senior vs Junior Doctors

Observation

Dermatology

- Total Doctors: 3
- Senior: 2
- Junior: 1

Pediatrics

- Total Doctors: 5
- Senior: 5
- Junior: 0

Oncology

- Total Doctors: 2
- Senior: 2
- Junior: 0

Business Insight Most departments rely heavily on senior doctors with limited succession planning through junior staff.

Recommendation Recruit additional junior doctors, especially in Pediatrics and Oncology, to ensure long-term workforce sustainability.

Doctor Workload Analysis

Observation Sarah Taylor handled the highest number of appointments (29), while Robert Davis managed the fewest (13).

Business Insight Doctor workload is unevenly distributed, which may lead to resource imbalance and increased workload for some physicians.

Recommendation Implement appointment balancing strategies to distribute patient load more evenly across available doctors.

4. Appointment Analysis

Appointment Status

Status	Count
No Show	52
Scheduled	51
Cancelled	51
Completed	46

Business Insight Completed appointments are lower than scheduled appointments, while no-shows and cancellations remain relatively high.

Recommendation Introduce automated appointment reminders, confirmation messages, and follow-up calls to reduce missed appointments.

Last 7 Days Activity

Observation Only **4 appointments** were recorded during the last seven days.

Business Insight Recent appointment activity is relatively low compared to historical records.

Recommendation Review appointment booking trends and investigate whether seasonal demand or operational factors affected patient visits.

5. Treatment Analysis

Most Common Treatments

Treatment	Count
Chemotherapy	49
X-Ray	42
ECG	38
MRI	36
Physiotherapy	36

Business Insight Chemotherapy is the most frequently performed treatment, indicating high oncology service utilization.

Recommendation Ensure adequate medical resources, equipment, and staffing for high-demand treatments.

Treatment Cost Analysis

- Minimum Cost: **534**
- Maximum Cost: **4,973**
- Average Cost: **2,756**

Business Insight Treatment costs vary significantly depending on the procedure performed.

Recommendation Regularly review treatment pricing and ensure transparency for patients regarding expected medical expenses.

Revenue Contribution by Treatment

Treatment	Revenue Contribution
Chemotherapy	23.38%
MRI	21.06%
X-Ray	20.07%
Physiotherapy	18.04%
ECG	17.46%

Business Insight Chemotherapy contributes the largest share of treatment revenue, followed by MRI and X-Ray services.

Recommendation Prioritize investment in high-revenue treatment services while maintaining quality standards.

6. Billing & Revenue Analysis

Payment Status Distribution

Status	Count
Pending	69
Failed	67
Paid	64

Business Insight Pending and failed payments exceed successfully completed payments, indicating inefficiencies in the billing process.

Recommendation Strengthen payment reminder systems, simplify payment methods, and investigate the reasons for failed transactions.

Total Revenue

Observation The hospital generated **173,424** in total revenue.

Business Insight The current revenue reflects the hospital's financial performance across all recorded billing transactions.

Recommendation Monitor monthly revenue trends and identify opportunities to improve revenue through operational efficiency and patient retention.

Top 10% Highest Spending Patients

Top contributors include:

- Laura Davis (P012)
- David Moore (P049)
- Emily Miller (P048)
- David Wilson (P035)

Business Insight A small group of patients contributes a significant share of total revenue, indicating the presence of high-value patients.

Recommendation Develop personalized follow-up programs, preventive healthcare packages, and loyalty initiatives to improve long-term patient retention.

Overall Business Recommendations

- Increase patient acquisition through community outreach and digital marketing.
 - Balance doctor workload across departments.
 - Recruit additional junior doctors for long-term workforce planning.
 - Reduce appointment cancellations using automated reminder systems.
 - Invest more resources in high-demand treatments such as Chemotherapy and Pediatrics.
 - Improve billing efficiency by reducing pending and failed payments.
 - Track key performance indicators (KPIs) monthly using interactive dashboards.
 - Build a Power BI dashboard for real-time monitoring of operational and financial performance.
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Conclusion

This Healthcare Analytics project demonstrates how SQL can be used to transform raw healthcare data into actionable business intelligence. The analysis identified trends in patient demographics, doctor utilization, appointment management, treatment performance, and revenue generation.

The findings highlight opportunities to improve operational efficiency, optimize healthcare resources, strengthen financial performance, and enhance patient satisfaction. These insights can support hospital

management in making data-driven decisions and provide a strong foundation for future business intelligence solutions such as Power BI dashboards and predictive healthcare analytics.